ChronoGate

Standard Operating Procedure — interactive time-gating, lifetime & phasor FLIM analysis

Version 0.18.0

2026-07-21

PicoQuant .ptu · Becker & Hickl .sdt

1. Purpose & scope

This procedure describes how to operate **ChronoGate**, an interactive time-gating, rapid-lifetime and phasor viewer for PicoQuant `.ptu` (and Becker & Hickl `.sdt`) FLIM data. It covers loading data, setting the time gate, producing lifetime and phasor maps, running a rigorous IRF-reconvolution fit, and exporting results.

It is a practical operating guide, not a validation report. Absolute-lifetime work should use a measured IRF (see §10) and be cross-checked against a reference of known lifetime.

2. What ChronoGate does

A pulsed laser excites the sample; for every detected photon the hardware records the delay since the last pulse (the **microtime**). The per-pixel histogram of those delays is the **fluorescence decay**. ChronoGate reconstructs that decay cube and lets you:

- **Time-gate** — keep photons in a chosen microtime window and sum them into an image (fit-free lifetime *contrast*).
- **RLD** — read an apparent lifetime from two gates (fit-free, per pixel).
- **Phasor** — plot each pixel's decay as a point on the universal semicircle.
- **IRF fit** — recover a rigorous, IRF-deconvolved lifetime by reconvolution.

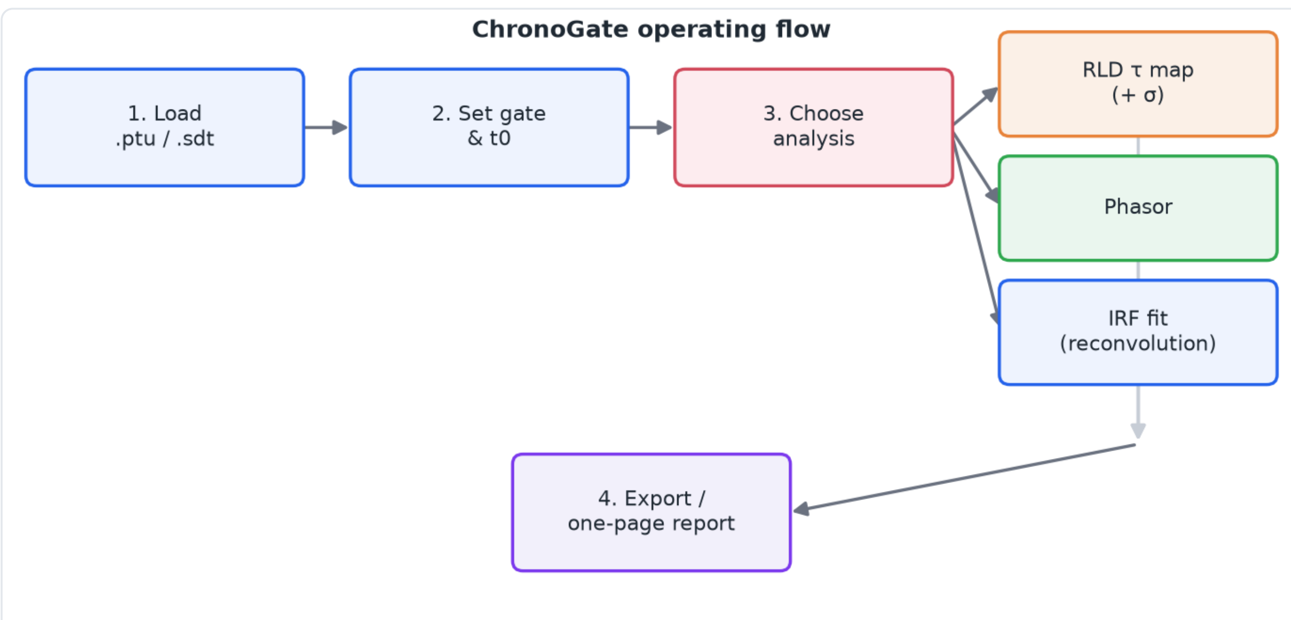

Figure 1. The end-to-end operating flow: load → gate → choose an analysis → export.

3. Install & launch

ChronoGate runs from source in a Python **3.12+** environment.

macOS (double-click launcher). Keep `ChronoGate.app` and `ChronoGate.command` together in the project folder and double-click `ChronoGate.app`. The first launch builds a virtual environment from `pyproject.toml` (a one-time download); later launches are instant. Changing dependencies triggers an automatic rebuild.

From a terminal (any OS).

```
python -m chronogate                    # opens a file picker
python -m chronogate path/to/file.ptu
python -m chronogate path/to/file.sdt
```

> If opening a `.sdt` reports that `sdtfile` is missing, or an IRF fit reports `scipy` is missing, your environment predates those dependencies — relaunch so the launcher rebuilds it, or run `pip install -e .` .

4. Opening data

- **File ‣ Open** (`Ctrl+O`) — choose a `.ptu` or `.sdt` file.
- **File ‣ Open folder (stack)** — load a numbered z-series (e.g. `FLIM_stack_z1.ptu … z65.ptu`); step through planes with the **z-slice** slider in the File panel.
- **Channel** — for multi-detector files, pick the detector channel.

5. The workspace

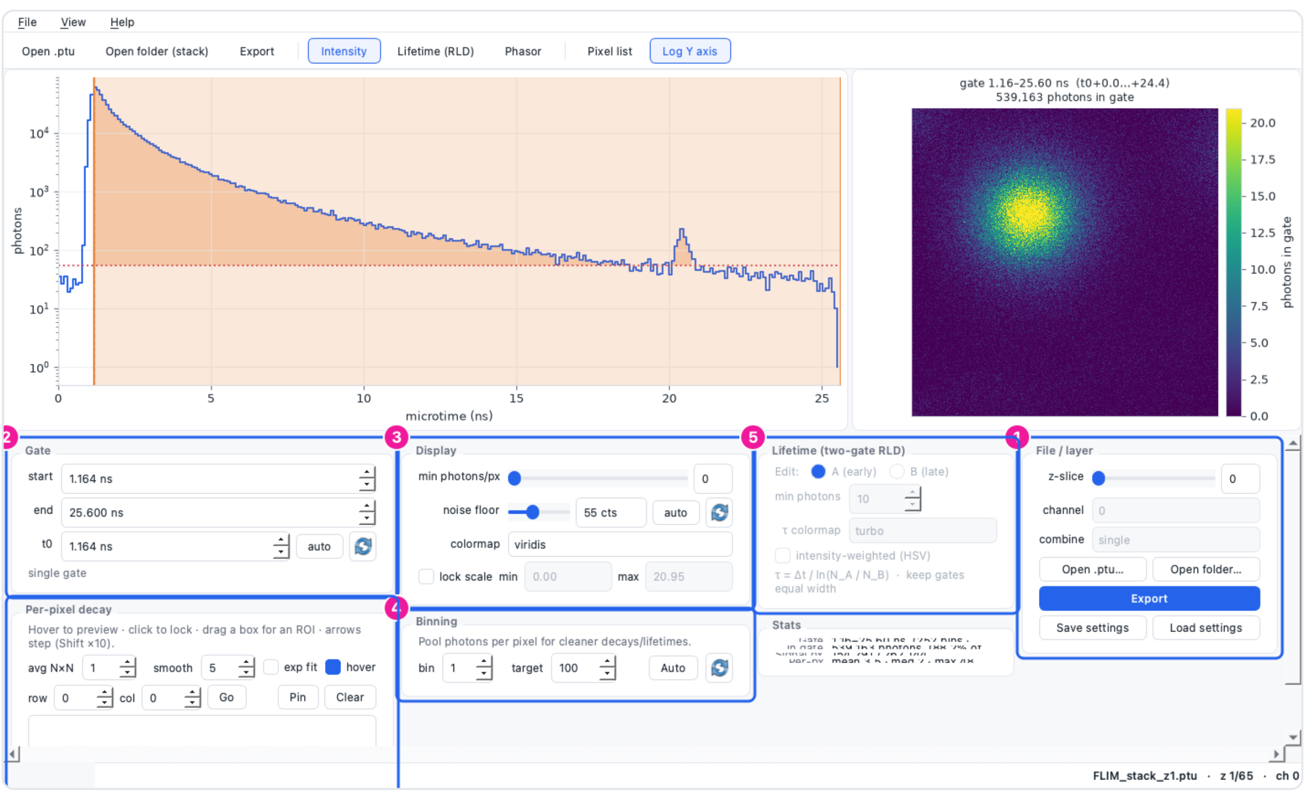

Figure 2. The main window. **1** File/layer (open, z-slice, channel, export, settings). **2** Gate (start, end, t0). **3** Display (threshold, noise floor, colormap, scale lock). **4** Binning. **5** Lifetime (RLD) controls. **6** Per-pixel decay / selection tools. The left plot is the summed decay with the gate shaded; the right plot is the current image.

Switch the image with the mode buttons in the toolbar: **Intensity** (`I`), **Lifetime (RLD)** (`T`), **Phasor** (`P`).

6. Setting the gate and t0

Figure 3. Gate panel: **1** gate start, **2** gate end (both in ns; you can also drag the shaded span on the decay plot), **3** t0 (pulse reference), **4** *auto* (set t0 to the smoothed decay peak), **5** *reset* (set t0 to its default, 0 ns).

1. Drag the shaded band on the decay plot, or type **start**/**end** in ns.
2. Set **t0** to the pulse arrival: click **auto** for the detected peak, or type a value. The **reset** (↻) button returns t0 to 0 ns.
3. The image updates live as you move the gate.

7. Display controls

Figure 4. Display panel: **1** min photons/px (blank dim pixels), **2** noise floor (counts, read on the summed decay), **3** floor *auto* (robust baseline), **4** floor *reset* (0 — no subtraction), **5** colormap.

- **min photons/px** masks pixels too dim to trust.
- **noise floor** subtracts a flat background so a pedestal doesn't bias the lifetime. **auto** picks a robust baseline; **reset** turns subtraction off.
- **lock scale** freezes the colour range so planes/frames are comparable.

8. Spatial binning

Figure 5. Binning panel: **1** bin factor, **2** target photons/px, **3** *Auto* (suggest a factor from photon statistics), **4** *reset* (1×, off).

FLIM pixels are often photon-starved. Binning pools each pixel's neighbourhood (`b×b`) for cleaner decays and lifetimes. Click **Auto** to reach the target photon budget, or set the factor by hand. **reset** (↻) returns to 1× (off).

> Binning is the fastest way to make a per-pixel lifetime *map* usable on sparse data — see §9 and §10.

9. Lifetime & phasor analysis

9.1 Two-gate RLD

Press **T** for Lifetime mode. RLD reads an apparent lifetime from two equal-width gates, per pixel, with no fitting:

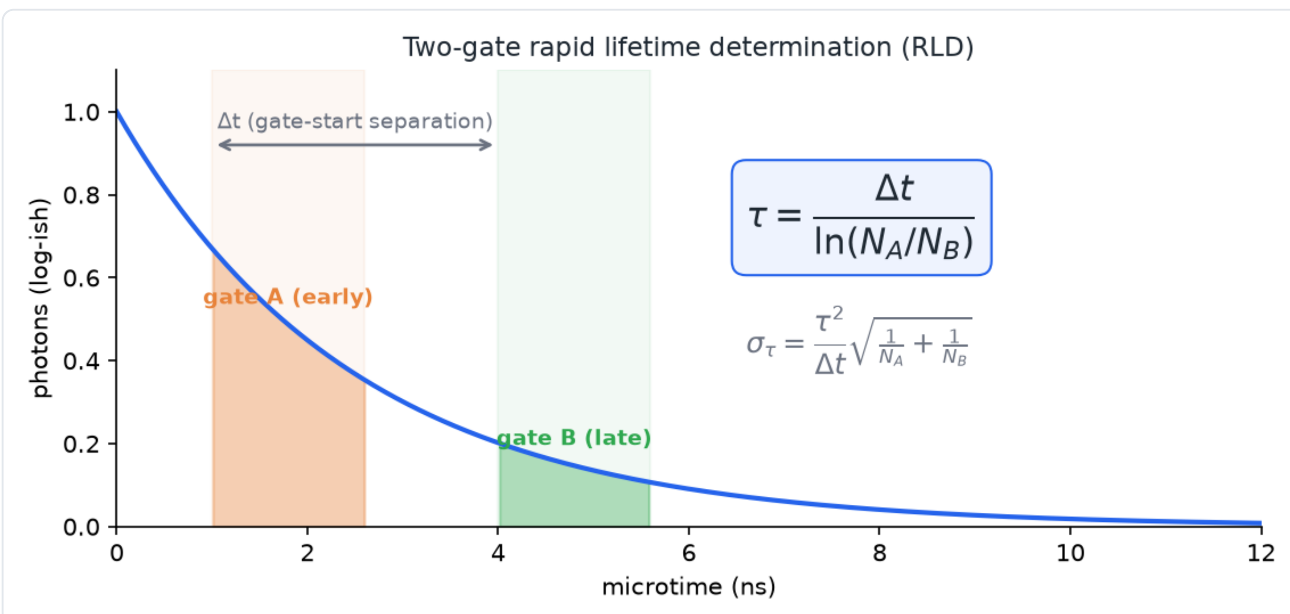

Figure 6. Two-gate RLD. With equal-width gates separated by Δt, the width/amplitude factors cancel, giving τ = Δt / ln(N_A/N_B). The shot-noise uncertainty σ_τ is reported for a pooled region.

1. In the Lifetime panel choose which gate you are editing (**A** early / **B** late) and position the two gates on the decay.
2. Keep the gates **equal width** (the panel warns if not) — RLD assumes it.
3. The τ image appears on the right. Select a region (§11) and open the one-page report (§12) to get the pooled **τ ± σ** (shot-noise uncertainty).

9.2 Phasor

Press **P**. Each pixel's decay becomes a point `(g, s)`; a single lifetime lands on the universal semicircle, mixtures fall inside it. Use **Calibrate phasor from reference…** (View menu) with a dye of known lifetime to remove the instrument/t0 offset, and the **2nd harmonic** toggle to spread short lifetimes apart.

10. IRF reconvolution fit (rigorous lifetime)

For an IRF-deconvolved lifetime (rather than fit-free contrast), use **View ‣ IRF lifetime fit…**.

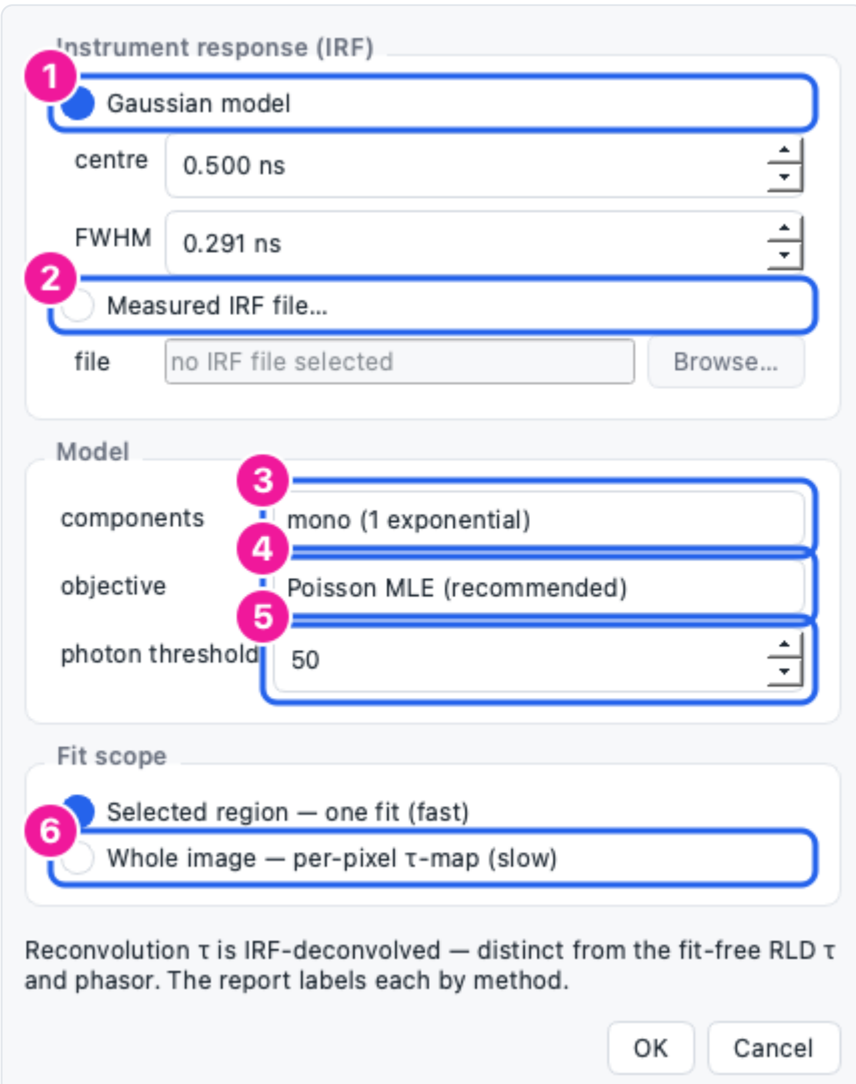

Figure 7. IRF fit dialog: **1** Gaussian IRF model, **2** measured IRF file, **3** components (mono/bi), **4** objective (Poisson MLE / weighted χ^2), **5** photon threshold, **6** whole-image τ-map (vs. a region fit).

1. **Choose the IRF.** For a first look, a **Gaussian** model is fine. For **absolute** lifetimes, load a **measured IRF** file (a scatter/reflection or reference-dye acquisition) — reconvolution is only as good as its IRF.
2. **Model** — mono for one lifetime, bi for two.
3. **Objective** — Poisson MLE (recommended for low counts) or weighted χ^2.

4. **Scope** — a **region fit** (select a bright ROI first) gives one trustworthy τ ± σ; a **whole-image τ-map** fits every pixel above the photon threshold.
5. Read the result: **reduced $\chi^2 \approx 1$** means a good fit. A large χ^2 (e.g. tens or hundreds) means the model is wrong — try a measured IRF, or bi-exponential (see Troubleshooting). A component reported as **σ = n/a (unidentifiable)** is an honest flag that the data cannot separate it.

> For a per-pixel map on sparse data, **bin first** (§8) so pixels clear the photon threshold, then run the map. The default threshold adapts to the image.

11. Selecting pixels & regions

- **Drag a box** on the image for a rectangular ROI; **lasso** on the phasor plot to select a population; or use the **Pixel list** (`View ▸ Pixel list`) to pick by metric.
- Hover to preview a pixel's decay; click to lock it; **Pin** to keep several for comparison.
- A selection travels with every export and scopes the report's statistics.

12. Exporting & the one-page report

Figure 8. Export dialog: **1** raster TIFF (raw values, for ImageJ), **2** one-page report (PNG+PDF with the headline number, decay, image and provenance), **3** restrict to the current selection, **4** *Export & open in Fiji*.

- **File ‣ Export** (`Ctrl+E`) — choose artefacts (raw TIFF, colormapped PNG, decay CSV, per-pixel table, selection files) and a folder. A content-hashed **provenance JSON** is always written.
- **File ‣ Export report** (`Ctrl+R`) — the one-page summary; in Lifetime mode it includes the pooled **τ ± σ** for the selection.
- **Restrict to selection** masks everything outside the selected pixels.

- **Export & open in Fiji** writes a raw TIFF plus an ImageJ macro (range, LUT, ROI) and launches Fiji on it (set the Fiji path in Preferences first).

13. Troubleshooting

Symptom	Cause	Fix
τ-map fits **0 px**	photon threshold above every pixel's count	bin more (§8), lower the threshold, or do a region fit
IRF fit **reduced $\chi^2 \approx$ 100s**	wrong/guessed IRF, or model too simple	load a measured IRF; try bi-exponential; check the Gaussian centre/FWHM
σ shows **n/a (unidentifiable)**	near-degenerate fit (e.g. two equal τ)	expected/honest — simplify the model or accept the point estimate
`.sdt` won't open (`sdtfile` missing)	environment predates the dependency	relaunch (auto-rebuild) or `pip install -e .`
RLD τ looks wrong	unequal gates, or no background subtraction	make gates equal width; set the noise floor (§7)
app "unidentified developer" warning	installers are unsigned	right-click ‣ Open (macOS) / More info ‣ Run (Windows)

14. Revision history

The SOP is regenerated from the app: edit `docs/sop/sop.md` and `docs/sop/figures.py` , then run `python docs/sop/build.py` .

Version	Date	Notes
0.18.0	2026-07-21	Generated from ChronoGate 0.18.0.